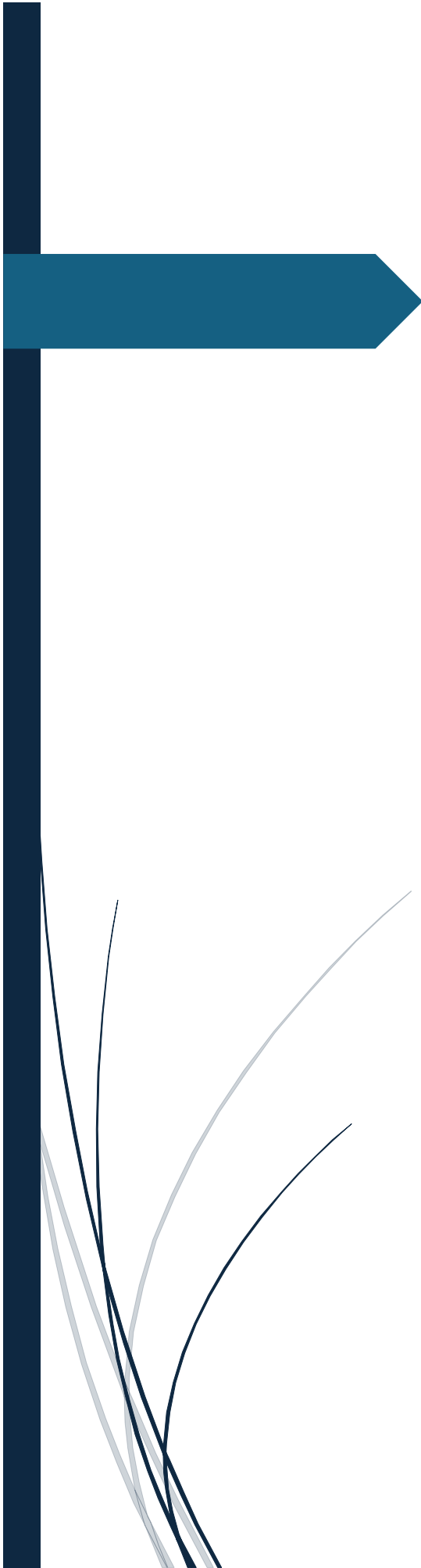




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Wave One

Complete Research Protocol: Combining SI, MSI, and Serum Lactate

1. Title of the Study

The Combined Prognostic Accuracy of Shock Index (SI), Modified Shock Index (MSI), and Initial Serum Lactate for Predicting In-Hospital Mortality in Polytrauma Patients.

2. Research Question: "Does the incorporation of Initial Serum Lactate with SI and MSI at Emergency Department (ED) triage enhance the early prediction of 30-day mortality in polytrauma patients relative to the use of individual markers?"

3. Background & Rationale

The Issue: Polytrauma is one of the death causes. It is important to identify Occult Shock early.

Biochemical Marker: Serum Lactate is a strong indicator of tissue hypoxia and anaerobic metabolism (Alshiakh, 2022).

Hemodynamic Markers: Shock Index (SI) and Modified Shock Index (MSI) are fast non-invasive early prognostication tools (Kim et al., 2022).

Synergy: The combination of these tools will give a complete picture of the circulatory and metabolic conditions of the patient.

4. Methodology

Study Design Prospective Observational Cohort Study.

Setting of the study: Level 1 Trauma Center / Emergency Department.

Sample Size: 200 Patients. (Based on the median sample sizes in Kim et al. and Alshiakh 2022).

Study Duration: 1 Month.

5.Participant Criteria

Inclusion: Adults (18 +), Polytrauma diagnosis and Return of Spontaneous Circulation (ROSC).

Exclusion: * Pregnancy.

Death on arrival.

Vital signs

absent or complete (Lactate).

Known pre-existing terminal illness (to avoid bias).

6. Clinical Cut-off Values

Shock Index (SI): 1.0mmol/L or more

Modified Shock Index (MSI): 1.3mmol/L or more

Serum Lactate: 2.0mmol

7. Data Collection Variables

Demographics: Age, Gender, Body Mass Index (BMI).

Comorbidities: Charlson Comorbidity Index (CCI).

Pre-hospital Data: EMS Response time, Bystander CPR, initial shockable rhythm.ED

Triage Vitals: Heart rate (HR), Systolic blood pressure (SBP), Diastolic blood pressure (DBP).

Computed Indices:

$SI = HR / SBP$

$MAP = (SBP + 2 \times DBP) / 3$

MSI = HR/MAP

Laboratory Data: First Arterial/Venous Lactate.

Outcomes: Repeat loss of circulation in ED, ICU length of stay, 30-day survival.

8. (Statistical Analysis)

"Following the methodology of Kim et al. (2022), we will use ROC curve analysis to determine the prognostic accuracy of SI and MSI. Cox regression will be used to estimate the Hazard Ratio (HR) of 30-day mortality, age and comorbidity (Charlson Comorbidity Index) adjusted.

9. Ethical Considerations

Information will be anonymized.

Informed consent will not be used (it will be an observational study of routine clinical data).

. References

1. **Kim, M. G. V. B., et al. (2022).** Prognostic value of the shock index and modified shock index in survivors of out-of-hospital cardiac arrest: A retrospective cohort study. *The American Journal of Emergency Medicine.*
2. **Alshiakh, S. M. (2022).** Association and prognostic value of blood lactate on different disease outcomes in the emergency department: A systematic review. *Journal of International Medical Research.*